

Report

Airline Passenger Analytics Dashboard (Power BI)

Project Overview

This repository features a robust Power BI dashboard designed to visualize and analyze airline passenger data. The dashboard uncovers valuable insights into passenger demographics, ticket revenue distribution, travel destinations, seat class preferences, and booking statuses. Both business and data teams can utilize this project to better understand travel patterns, improve sales strategies, and enhance passenger experience.

Data Sources

- **Passenger Manifest:** Contains personal and demographic details.
- **Ticket Pricing:** Provides fare information for each passenger.
- **Flight Details:** Includes flight numbers, origins, and destinations.

Data Cleaning and Preparation

Data Cleaning Steps

- **Duplicate Removal:** Ensured each PassengerID is unique by checking and removing any duplicate records to maintain individual-level analysis integrity.^[1]
- **Missing Value Handling:** Reviewed all columns for missing or null values. Where available, absent data was filled using logical inferences (e.g., if Age Group missing but age available, inferred group accordingly). Unusable records (with multiple critical fields missing) were excluded from analysis.^[1]
- **Data Type Correction:** Converted all numeric columns (such as TicketPrice, Age) to appropriate data types, and standardized categorical fields like Gender ("Male", "Female") and Status ("Confirmed", "On Hold", "Cancelled") for seamless visual aggregation.^[1]

- **Outlier Detection:** Inspected Ticket Price and Age fields for anomalies (e.g., extraordinarily high prices or implausible ages) and validated accuracy against other attributes wherever possible.

Data Transformation Steps

- **Age Group Categorization:** Classified passengers into age groups (18-25, 25-35, 35-45, 45-50) using the Age attribute, supporting segment-based visualizations and reports.^[1]
- **Status Label Normalization:** Streamlined values in the Status column, unifying variants (e.g., "Confirm", "confirmed", "On Hold", "Hold") to a consistent format ("Confirmed", "On Hold"). This allows correct slice-and-dice reporting.
- **Seat Class Grouping:** Standardized the Seat Class entries and ensured the main classes ("Business", "Economy", "First") appear uniformly for cross-passenger revenue comparison.
- **Source-Destination Pairing:** Parsed Departure and Destination columns for clarity, enabling meaningful geographic route analysis by mapping valid pairs.
- **Revenue Aggregation:** Created derived fields such as total revenue per Seat Class and passenger, useful for bar and trend charts.
- **Status Count Calculation:** Aggregated booking status counts for gender, age group, and other sub-dimensions, supporting pie and bar chart summaries on the dashboard.

Dashboard Features and Visualizations

Key Metrics and Tiles

- **Total Ticket Revenue & Passenger Count:** Instantly view total ticket revenue and passenger headcount via prominent KPI cards.
- **Interactive Filters:** Analyze subsets of data by filtering on Source, Destination, Gender, Status, and Age Group.

Visual Insights

- **Ticket Price by Passenger:** Line chart illustrates ticket price variability across individual passenger IDs, highlighting both outliers and trends.
- **Class-wise Revenue:** Bar chart delineates total revenue generated from Business, Economy, and First-Class seat bookings.
- **Gender and Age Group Distribution:** Pie and bar charts depict passenger splits by gender and age segment, aiding customer targeting and campaign planning.
- **Booking Status Analysis:** Pie chart visualizes breakdown of Confirmed, On Hold, and Cancelled bookings, identifying potential bottlenecks and sales opportunities.

Tabular Data Model

The underlying data table contains:

- Passenger ID, Full Name
- Age, Gender, Age Group
- Flight Number, Departure, Destination
- Ticket Price, Seat Class, Booking Status

Customized calculated columns and measures (created via Power BI's DAX language) drive most summarizations and aggregations.

Analytical Insights

- **Predominant Travelers:** Ages 25–35 comprise the most frequent flyers.
- **Gender Breakdown:** Female passengers form a larger sample in this data.
- **Revenue Drivers:** Business class yields the highest combined ticket value.
- **Booking Pipeline:** “Confirmed” bookings are the majority, but “On Hold” and “Cancelled” highlight areas for optimizing the booking funnel.